F THE FLOW OF AUTOREF-HARNESS



Figure 8: **Execution trajectory of AutoRef-Harness on a held-out four-reference task.** Steps 1–5 show the input, the identification of the scene reference, the two drafts, failure-aware selection, and complaint-directed revision. Quoted text with a blue bar is a prompt sent to the reasoning model M_θ, and the text below it is the model's response; boxes labeled "image model" mark calls to the generator G_ϕ. Prompts and responses are abridged but not otherwise modified. Each reference's frame color matches that of its number in the text.

Figure 8 shows how AutoRef-Harness processes one task. The reasoning model first infers that image 2, the poodle in a library, sets the scene (step 2). It then writes two structurally different prompts (step 3): a scene-first prompt that describes the setting and then each subject (draft A), and a canvas-anchored prompt that passes image 2 first and places the other subjects in it (draft B). Both drafts pass the failure check, and draft B is preferred in both presentation orders (step 4). The reasoning model lists complaints about draft B, mainly that the violinist in image 4 does not match